

PC Cafe Simulation

Cybercafe capacity planning and SimPy simulation documentation

Sanitized portfolio copy. Original school branding, cover-page personal details, metadata, and live PDF links were removed. This is not an official university publication.

1. INTRODUCTION OF THE PROJECT

A local cybercafe case context was used for the simulation model.

h a

finite capacity of 56 PC units. Operating under non-stationary demand, the experiences significant traffic spikes during peak hours, weekends, and evening operations. This project implements a Discrete-Event Simulation (DES) to model hardware utilization, customer queueing, and reneging behaviors, enabling data-driven capacity planning without relying on sensitive internal customer data.

2. PROBLEM DEFINITION

The facility faces capacity challenges when customer arrivals exceed the available 56 PC units. Currently, customers wait in the cafe if units are occupied, but they do not wait indefinitely; observed behavior indicates that customers typically abandon the queue after 5 to 10 minutes. Given the complexity of the account-based top-up system, early departures, and group behaviors, manual observation is insufficient to determine true capacity pressure. The primary problem is hardware starvation and lost demand during peak periods.

3. OBJECTIVES OF THE SIMULATION

This study aims to model the operational efficiency of PC unit usage at through the following objectives:

1. **Identify Hardware Starvation:** Determine the probability and duration of the system reaching 100% PC utilization.
2. **Evaluate Queueing Dynamics:** Measure average wait times, queue lengths, and occupancy at capacity.
3. **Analyze Reneging Behavior:** Estimate customer abandonment rates based on wait tolerance.

4. **Compare Solo and Group Behavior:** Observe how segment-specific behavior affects PC allocation and wait times.
5. **Evaluate Capacity Expansion:** Assess the impact of adding PC units on throughput, utilization, and reneging.
6. **Estimate Lost Demand:** Quantify potential demand loss due to unavailable resources.

4. SYSTEM MODEL AND DESIGN

Identified Entities, Events, and Resources

Entities (Dynamic)

- **Customers:** The primary dynamic entities in the simulation. Each customer has stochastic attributes such as `Customer\_Type`, `Session\_Duration`, `Tolerance\_Wait\_Time`, `Top\_Up\_Decision`, and `Early\_Logout\_Status`.
- **Solo Customers:** Individual customers who require one PC unit. They represent approximately 70% of customer arrivals. They are expected to have slightly higher queue tolerance than groups.
- **Group Customers:** Customers arriving together as a group. They represent approximately 30% of customer arrivals. They are more likely to abandon the queue faster if they cannot be seated near each other.

Resources (Static/Constrained)

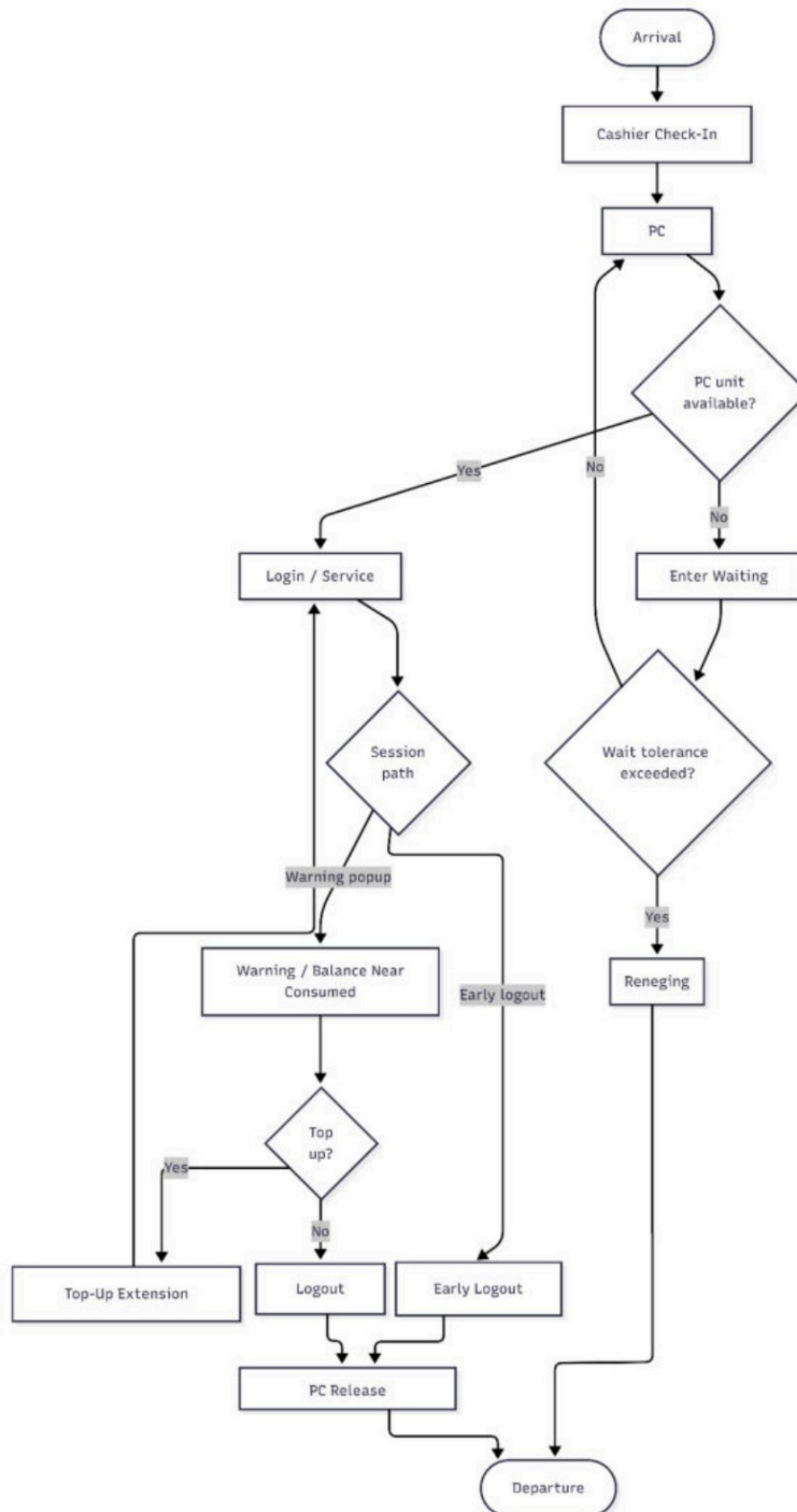
- **PC Units (Nodes):** The primary constrained resource of the system. The base model starts with 56 concrete PC unit IDs, and capacity-expansion scenarios can add simulated PC units. The simulation tracks which exact unit IDs are occupied, released, or added.

- **Waiting Area:** The indoor cafe space where customers wait when all PC units are occupied. The waiting area is modeled as a logical queue rather than a separate physical resource.
- **Cashier/Top-Up Interaction:** A visual-only logical event point for customer check-in, top-up, and exit activity. It is not modeled as a constrained cashier queue because cashier service time and cashier records are outside the study scope.

Events (Discrete)

- **Arrival:** A customer or group enters the cybercafe and requests one or more available PC units.
- **Cashier Check-In:** The arriving customer passes through the cashier as a visual-only step before requesting a PC unit. This event does not add service delay.
- **PC Request:** The customer attempts to acquire a PC unit. If a PC is available, the customer proceeds to login. If all PCs are occupied, the customer enters the waiting queue.
- **Login (Service Start):** The customer occupies a PC unit and begins active computer usage.
- **Warning Popup:** The system indicates that the customer's remaining balance or available time is nearly consumed.
- **Top-Up Extension:** A customer adds more time to their account and continues occupying the PC unit.
- **Early Logout:** A customer logs out and leaves before fully consuming their available account balance (occurs around 60% of the time).
- **Logout (Service Completion):** The customer finishes using the PC.
- **Departure:** The customer vacates the premises.
- **Reneging:** A waiting customer leaves the queue before receiving a PC because their tolerance wait time has been exceeded.

The simulation models the lifecycle of a customer visit through the following required process flow:



5. ASSUMPTIONS AND INPUT DATA

Initial Assumptions

1. **Arrival Pattern:** Non-stationary Poisson process based on time of day and day type.
2. **Customer Composition:** 70% solo customers and 30% group arrivals.
3. **Session Duration:** Derived from account-based top-up system usage; customers may leave early.
4. **Top-Up Behavior:** Visual-only cashier interaction; does not create a separate service queue.
5. **Queue Discipline:** Visual check-in at cashier followed by PC request.
6. **Wait Tolerance:** 510 minutes before renegeing; groups have shorter tolerance.
7. **Group Seating:** Conceptual same-row seating constraint for groups.
8. **Hardware State:** All 56 units treated as operational.

Scenario Presets and Adjustable Parameters

The model utilizes scenario presets to compare operating conditions. Users may adjust: arrival multipliers, top-up probabilities, early logout probability, group seating configurations, and added PC unit capacity.

6. SIMULATION IMPLEMENTATION AND DEMONSTRATION

The simulation is implemented as follows:

1. **Engine:** Discrete-event simulation developed in Python using the SimPy library.
2. **Source Files:** Jupyter Notebook containing EDA, scenario logs, charts, and metrics.
3. **Deployment:** Gradio dashboard for Hugging Face Spaces.
4. **Features:** Real-time visualization of queue states, per-unit elapsed timers, and capacity-expansion controls.

7. EXPERIMENTAL RESULTS AND ANALYSIS

The following results were extracted from 10 replications per main scenario.

Scenario Performance

- **Baseline Weekday:** Avg. Utilization 86.493%, Reneging 37.683%.
- **Weekend High Demand:** Avg. Utilization 92.715%, Reneging 55.55%.
- **Group Seating Sensitivity:** Avg. Utilization 84.999%, Reneging 38.545%.
- **Top-Up Extension:** Avg. Utilization 86.993%, Reneging 41.225%.

Capacity Expansion Comparison

Setup	PC Count	Added Units	Avg Wait Time	Throughput/hr	Utilization	Reneging
Baseline Weekday	56	0	1.879 min	13.636	86.493%	37.683%
Baseline Weekday + 8 PCs	64	8	1.445 min	15.082	84.705%	29.503%

The analysis confirms that adding 8 simulated PC units reduces reneging from 37.683% to 29.503% and improves overall throughput.

8. CONCLUSION AND RECOMMENDATIONS

The discrete-event simulation demonstrates that the cybercafe is constrained during peak hours. Reneging is highly correlated with wait tolerance, particularly among group customers. It is recommended to expand capacity based on the simulated 64-PC setup to reduce demand loss.

REFERENCES

- [1] E. C. Salas, M. Arias, J. D. Rodríguez, M. B. Silva, A. Solís, C. Fernandez, S. Aguirre, and E. Rojas, "PROcess Mining and Simulation for Healthcare Analysis: PROMSHA-Methodology," 2026, doi: 10.1109/access.2026.3663154.
- [2] T. Benarbia, H. A. Ahmad, A. K. Kayisu, J. C. Chedjou, D. J. Sun, and K. Kyamakya, "Behavior Modeling and Performance Assessment of E-Scooter Rental Mobility Systems in Discrete Event Simulation Framework," 2026, doi: 10.1109/access.2026.3668018.
- [3] C.-W. Kim, S.-H. Ju, and K.-M. Seo, "Discrete-Event Simulation Framework for Composite Warfare With Modular CFCS Modeling for Naval Combat System," 2025, doi: 10.1109/access.2025.3648783.
- [4] J. P. Bertucci, M. Salazar, and T. Hofman, "Battery Electric Truck Infrastructure Co-Design via Joint Optimization and Agent-Based Simulation," 2025, doi: 10.1109/tte.2025.3575704.
- [5] M. G. de Alcantara, A. F. da Silva, F. A. S. Marins, and R. C. Carvalho, "Systematic Literature Review on Multi-Objective Simulation Optimization," 2025, doi: 10.23919/csms.2024.0037.
- [6] B. Soykan, G. Blanc, and G. Rabadi, "A Proof-of-Concept Digital Twin for Real-Time Simulation: Leveraging a Model-Based Systems Engineering Approach," 2025, doi: 10.1109/access.2025.3557367.
- [7] K. A. Haque, S. Sun, X. Huo, A. Goulart, and K. Davis, "Scalable Discrete Event Simulation Tool for Large-Scale Cyber-Physical Energy Systems: Advancing System Efficiency and Scalability," 2025, doi: 10.1109/access.2025.3578948.
- [8] S. Zhang, J. Li, C. Liu, L. Fu, Z. Bai, and J. Li, "SIGMA: An Agent-Based Modeling UAV Swarm Simulator for Swarm Intelligence Algorithms," 2025, doi: 10.1109/tase.2025.3597077.
- [9] G. Li, L. Zhang, and X. Zhao, "Discrete Event Simulation-Driven Method Solving Permutation Flowshop Scheduling Problem in Digital Twins," 2025, doi: 10.1109/access.2025.3541223.
- [10] I. Franco-Alucano, G. Tortorella, J. Aguilar-Duque, D.-L. Flores, J. Limón-Romero, Y. Báez-López, and D. Tlapa, "Lean Healthcare 4.0: Integrating AI, Lean Principles, and Simulation for Laboratory Process Improvement," 2025, doi: 10.1109/access.2025.3640970.